

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 7

Question 1 - MCQ · PG-16 · LR-1

A set which does not contain any element is called an empty set.
Which symbol denotes the empty set?

Options

- A. $\emptyset$
- B. $\{0\}$
- C. $\{\emptyset\}$
- D. 0

Correct answer:

Explanation and steps

An empty set has no elements. Its standard symbol is $\emptyset$; $\{\emptyset\}$ is a set containing one element.

Question 2 - MCQ · PG-16 · LR-2

Complete the distributive property.

$$A \cup (B \cap C) =$$

Options

- A. $(A \cup B) \cap (A \cup C)$
- B. $(A \cup B) \cup (A \cup C)$
- C. $(A \cap B) \cup (A \cap C)$
- D. $(A \cap B) \cap (A \cap C)$

Correct answer: $(A \cup B) \cap (A \cup C)$

Explanation and steps

Union distributes over intersection: combine A with each set by union, then take the intersection.

Question 3 - MCQ · PG-16 · LR-3

Complete the distributive property.

$$A \cap (B \cup C) =$$

Options

- A. $(A \cap B) \cup (A \cap C)$
- B. $(A \cap B) \cap (A \cap C)$
- C. $(A \cup B) \cap (A \cup C)$
- D. $(A \cup B) \cup (A \cup C)$

Correct answer: $(A \cap B) \cup (A \cap C)$

Explanation and steps

Intersection distributes over union: intersect A with each set, then take the union.

Question 4 - MCQ · PG-16 · LR-4

Complete De Morgan's law.

$$(A \cup B)' =$$

Options

A. $A' \cap B'$

B. $A' \cup B'$

C. $A \cap B$

D. $A \cup B$

Correct answer: $A' \cap B'$

Explanation and steps

By De Morgan's law, the complement of a union equals the intersection of the complements.

Question 5 - MCQ · PG-16 · LR-5

If A and B are finite disjoint sets, then $n(A \cup B) =$

Options

A. $n(A) + n(B)$

B. $n(A) + n(B) - n(A \cap B)$

C. $n(A) - n(B)$

D. $n(A) \times n(B)$

Correct answer: $n(A) + n(B)$

Explanation and steps

Disjoint sets have no common element, so $n(A \cap B) = 0$ and no subtraction is required.

Question 6 - MCQ · PG-16 · LR-6

If A and B are finite sets and $A \cap B \neq \emptyset$, then $n(A \cup B) =$

Options

A. $n(A) + n(B) - n(A \cap B)$

B. $n(A) + n(B)$

C. $n(A \cap B) - n(A) - n(B)$

D. $n(A) \times n(B)$

Correct answer: $n(A) + n(B) - n(A \cap B)$

Explanation and steps

Adding $n(A)$ and $n(B)$ counts their common elements twice, so subtract $n(A \cap B)$ once.

Question 7 - MCQ · PG-16 · LR-7

Complete the Cartesian-product result.

$R \times R =$

Options

A. $\{(x, y) / x, y \in R\}$

B. $\{x / x \in R\}$

C. $\{(x, y) / x \in R, y \in N\}$

D. $\{(x, x) / x \in R\}$

Correct answer: $\{(x, y) / x, y \in R\}$

Explanation and steps

A Cartesian product contains ordered pairs. In $R \times R$, both the first and second components are real numbers.